AI policy for MA theses in Applied Economics

- Based on AI policy by Gábor Békés (Central European University, BekesG@ceu.edu)

AI policy: Enjoy "AI" with moderation

We assume that throughout the program, you have developed core coding and statistics/econometrics skills, as well as the ability to read and understand economic research and implement your own design (under supervision). The abundance of AI means that the process of doing data analysis projects has changed, and these tools are becoming essential help.

AI has become part of our lives, and most people are using it regularly. **Therefore, we assume that you will use AI as a tool to enhance your work**. Moreover, the ability to use "AI" as a copilot, carry out prompt engineering is a new and important skill to learn and practice.

With this in mind, here are the main considerations:

1. **You are responsible** for all aspects of your work. AI is like advanced search; it's helpful, but use it as an input. You must carefully check and debug any input from AI. AI can help you in all the steps, but **you need strong foundations to become an applied economist who uses AI, but cannot be replaced by an LLM.**

2. **Policing the use of AI tools is already difficult and will soon be impossible.** Instead, your advisor and examiners will make sure that you can show in your communications with them that the work you present is your intellectual work. That means having a thorough understanding not only of what you have done but also of what you have not done. For instance, if you implement a certain empirical specification, you should be able to explain why other specifications were not considered.

3. **It is in your best interest to use AI as a tool and not as a substitute.** You have decided to spend substantial time and effort to acquire skills in our MA program. A crucial part of this process is the production of original work in your MA thesis. **Abusing AI tools means giving up on such skills.**

On certain specific tasks:

1. **Literature Review**

You may ask an LLM to help you find relevant papers, the same way that you may use Google Scholar. However, going through published papers is a necessary step before engaging in your own research. You should not cite papers you have not read or at least gone through yourself. Keep notes on the papers you plan to use and make sure you have a good idea of the question/setting/method they use and the contribution of your work relative to it.

Some suggestions: how can AI help

- Learn more about topics by having a conversation with AI on the material. You may copy text into it and ask for help in understanding details and clarifications.

- You may ask about regulations you are trying to study or exploit, about the construction of variables, or the design of a survey, etc.

- You may ask for help with derivations, math. For instance, if you are reading a paper and they have two equations and you cannot figure out how to get from the first to the second, LLMs can be a huge help. In our experience, LLMs can produce decent-quality math but need the right prompting. If you do not have a clear idea of what you want, LLMs will lead you astray.

2. **Presentations**

You are expected to understand the paper you present deeply and create the content of the presentation on your own. But you may use AI when crafting your presentation

Some suggestions: how can AI help

- Technical help re markdown/quarto or latex/beamer, etc formatting
- General help re formatting, simplifying slides
- Graphics, videos

3. **Writing**

The most natural path here is for you to write a first draft and then ask an LLM to offer suggestions to improve the writing, and it is up to you to decide which to implement. It may also help with structuring, etc., but this requires further sophistication.

If the whole text / large chunks is/are written by an LLM, your advisor will certainly notice. She may then be inclined to investigate further the level of involvement you had with the whole project. Therefore, I suggest you simply use the suggestions one by one and make sure the text remains yours.

LLMs have a very specific way of writing that you may already be familiar with. Humans do not write like that. In practice, your advisor may take

4. **Coding**

You may use any "AI" assistance for coding. Note that you carry all responsibility for the correctness of your code. Please test and debug your code relentlessly.

You are responsible for your code, no matter how you wrote the actual lines.

You will be expected to understand all aspects of your code and be able to explain it clearly.

Some suggestions: how can AI help

Drafting code snippets, turning code bits into functions,

- Improving code, commenting it
- Translate from one language to another, like Stata to R.
- Help create tables and graphs
- Debugging

5. **Data Analysis**

You shall design data wrangling and plan to do the analytical work first without AI. Decide on what the work data shall look like. Think about which methods you'd like to pursue and compare. Then comes AI, get help with actual work, find packages, ask for help with comparing tools. In many cases,

especially with common issues, AI will suggest great solutions. In other cases, such as graphics, AI will be great to get to about 80% of what you'd want.

In terms of discussing your results, AI will be very good in simpler tasks, such as descriptive statistics and good but not perfect in econometrics. It will be convincing, but often imprecise. Most importantly, relying on LLMs will not prepare you to explain the results to your examiners.

AI attribution statement

Please include an AI attribution statement. In it you shall list what AI tools you used, and in what capacity. You can use IBM's version, see the description here

Ideally, also keep a unique file with all the prompts and responses used during the process that can be consulted upon further request.